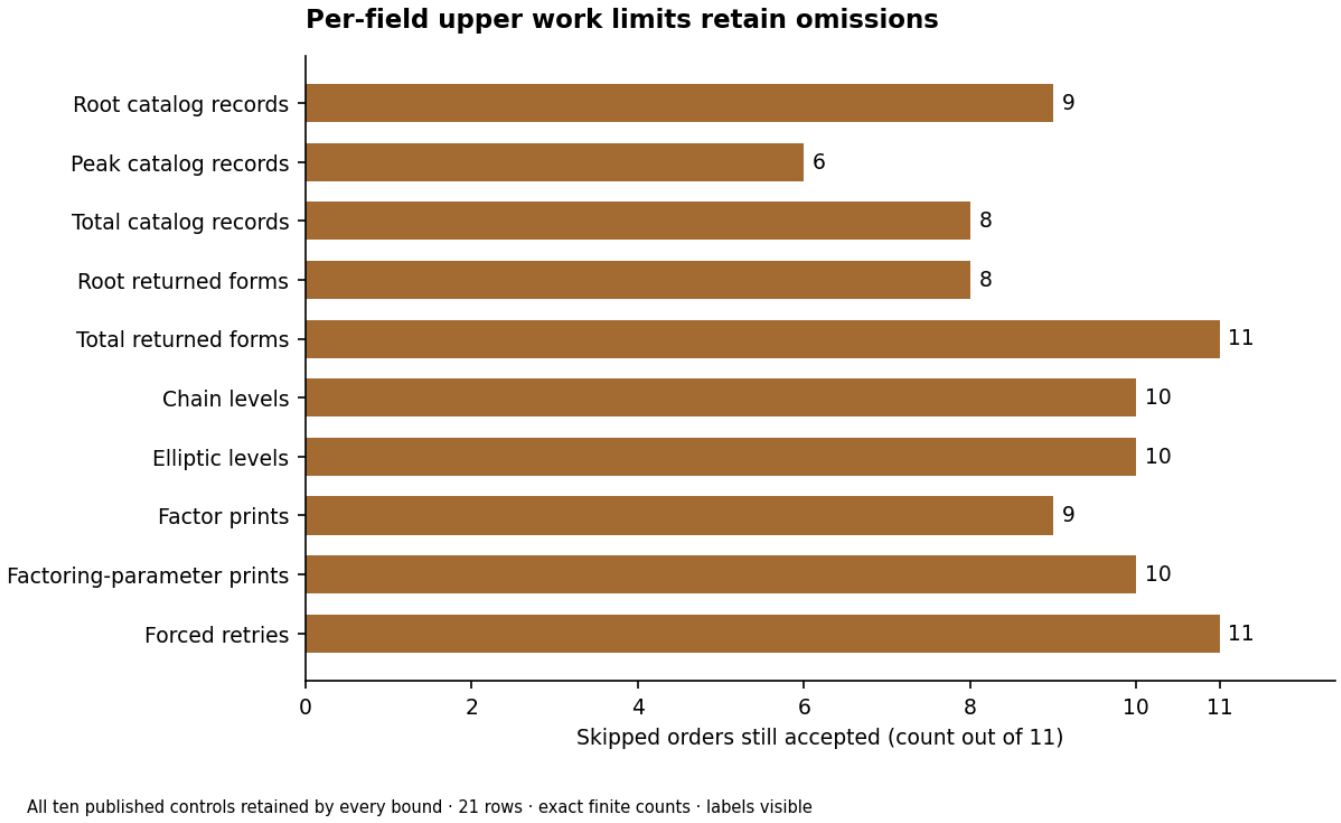
Recorded ECPP work does not explain every skipped curve

27 October 2026. All 21 saved order-proof transcripts were analyzed natively, retaining ten published and eleven omitted choices. All 210 predefined work counts are known. None of the 40 declared single-feature cutoff comparisons separates the choices. Even field-specific combined upper limits retain two omissions: p239 stream indices 1562 and 2682. **The historical selection criterion remains unresolved.** No vulnerability or measured DLP is established.



All ten published controls retained by every bound · 21 rows · exact finite counts · labels visible

Each bar is the number of omissions still accepted by the tightest per-field upper bound on that feature which keeps every published control. These are exact finite counts from the native comparison, with visible labels and no statistical interval or held-out prediction. Counts and stream indices are decimal. The complete table retains all 21 rows; exact p,b,n,seed identities preserve the original curve boundary. This order-only analysis introduces no curve aliases.

A stronger, checked counterexample

Both omitted p239 orders have no larger value in any of the ten recorded quantities than published prime239v1 (index 2185) and prime239v3 (index 3048). The separate token-scanning verifier re-extracted every count from original text and found all four dominance pairs. All 40 recomputed envelopes agree with the first implementation. Three altered-table controls are refused for the intended wrong-count, missing-row and changed-status reasons.

Exact count vectors: each omission ≤ each published control

	Root catalog	Peak catalog	Total catalog	Root forms	Total forms	Chain levels	Elliptic levels	Factor prints	Parameter prints	Forced retries
p239-1562 omitted	1	5	31	1	26	14	8	52	42	0
p239-2185 published	2	19	77	2	46	20	12	55	86	0
p239-2682 omitted	1	9	42	1	33	18	9	55	43	0
p239-3048 published	10	10	63	5	52	27	20	59	92	0

Native table → literal data bridge → Lean constants · no scores or normalized values

The table shows literal native-checked values, not normalized scores. For a cost C which is nondecreasing in these ten quantities, u≤v componentwise implies C(u)≤C(v). If the published vector v fits a budget B, transitivity gives C(u)≤C(v)≤B for the omitted vector u. Therefore an upper-budget screen on these counts cannot reject either omission while accepting prime239v1. This also constrains nonlinear monotone costs, beyond the single cutoffs or their conjunctions. It says nothing about additional unrecorded work or a different acceptance predicate.

The Lean source, literal data bridge and proof map make that conditional argument explicit. Lean 4.34.1 checks four exact vector inequalities, natural-valued and general transitive-cost implications, and the two-omission corollaries. The unchanged repository axiom auditor accepts eight theorems with empty transitive axiom lists: zero failures and zero trusted native evaluations. The data constants are bound to the independently checked table. Monotonicity and transitivity are model assumptions, not asserted properties of historical runtime. This is not a formalization of curve security or primality certificates.

What ran and what the quantities mean

The source-bound protocol and custom saved-analysis plan preceded complete-cohort extraction. The non-freeze plan hash is 85a646f6d5e87f90b2b7e96bf7c327fd4969ab2615de0805d7acc08729ce2d77. All labels and one example transcript's format were already visible. This is an analysis of the fixed census, not blinded preregistration. The additional dominance diagnostic and independent validation protocol were declared after the primary output; original source, plan and results remain unchanged.

One strict C++17/O2/Wall/Wextra/Werror invocation processed all 21 original replay004 packets under source b1f58e9d8e08bb3a9e1794166cd69e7556bbffa3. Saved verification under 27018452d455be75b27167812cb0d8f11a2876d4 uses a separate line/token implementation instead of the first parser's regular expressions. Both use a 60-second watchdog and 768 MiB address-space ceiling. The frozen source/input and binary checks pass before and after execution. All original ECPP and FreeLIP programs ran zero additional times. No wall/CPU/RSS clock measurement is admitted.

The ten quantities are root, peak and total catalog-record scans; root and total returned forms; build-chain levels; elliptic-route levels; printed factor records; printed factoring-parameter records; and forced retries. They concern proving the integer group order n prime. The original curve's point-counting workload and ECDLP cost are outside this vector. Printed factor records include repeated prints and are not counts of unique prime factors.

The original source audit matters: findD silently skips catalog records failing genus/norm tests, so next-D messages alone undercount catalog scans. In this rank-zero, serial configuration, the last returned discriminant's unique catalog rank gives records consumed at each depth when no retry/exhaustion occurs. All 21 traces satisfy those conditions; the independent parser checks increasing catalog ranks. Explicit N_i/D[[i]] pairs count the terminal build level, whose timing line can be omitted. The subsequent Pratt record is separate. This is recovered V3.4.1 source, not an authenticated original curve-generation caller.

Declared comparison	Observed result
Forty single-feature envelopes	Zero complete separators
All ten global upper bounds jointly	All 11 omissions retained
All ten per-field upper bounds jointly	Two omissions retained: p239-1562, p239-2682

Declared comparison	Observed result
Same-field pointwise dominance	Four pairs, using published indices 2185 and 3048
Historical attribution	Not established

No unknown row was dropped or encoded numerically. The old ECPP certificate admissions remain 3 published and 6 omitted accepted, with 7 published and 5 omitted indeterminate. All twelve unknowns stay text labels and enter no work-count predicate. Their mathematical group-order primality had already been checked separately in the earlier study. A complete work profile does not turn a failed legacy theorem transcript into a valid certificate.

Source review and preserved failures

The renewed review found a 24 October lab prefix reconstruction, retrieved at the immutable commit in upstream custody. It already reports 24 matching FreeLIP call/argument/curve paths, six matching manual-seven baselines and 24 nonmatching controls. Its portable-build and Python-verifier provenance is retained. No such Python code was executed here. The seven-call compatibility is prior reported work; historical invocation remains unproved. The corresponding period-NTL source audit describes a different default path. Neither modular RNG automatically shifts the separate lrand48 seed stream; coupling needs a caller.

The primary Anthropic report, Valsorda enumeration and Mattsson TLS message still leave the omissions unexplained. The retrieved problem record also reports broader 69/99-position draft-based cohorts. Those remain distinct from these 21 fixed records; no untested broader rule is certified here. An unpublished eligible seed position is not, by itself, proof of an original program visit followed by rejection. The original caller and point-count inputs are still missing from the retrieved evidence.

Proof development records preserve two thread-startup failures under a 4 GiB virtual-address cap, a decidability elaboration failure, an auditor memory failure and a PATH failure. Unfolding the unchanged predicate resolved decidability; no statement, vector or axiom was weakened. Final proof checking uses one worker and internal 1 GiB/4 GiB allocation limits for build and audit, without a virtual-address cap or measured RSS. The exact official release bytes match the published SHA256. lean4checker was unavailable. See the proof reproduction guide and descriptive run reports.

An intervening storage migration broke nine earlier file bindings. Exact original paths, metadata and PDF bytes were restored; all 97 bindings pass. The custody record preserves both observations without attributing the change to an unverified actor. Scientific bytes remain unchanged.

Scope and evidence custody

User requirement	Status
Four outstanding factorizations and full comparison	Preserved complete in the earlier factor follow-up; all 21 controls retained
Fixed ECPP work-limit explanation	Excluded for the declared limits and monotone costs of these ten counters
Historical reason for every skip	Unresolved; no supported complete criterion found
Selected/skipped weakness hypotheses	Neither established; no applicable attack or measured DLP

The preparation archive was fully downloaded and all 27 members checked before the primary invocation; the validator binary also had a full remote byte check before saved validation. The final archive has 144 verified member files and includes exact copies of all 65 consumed original packet/catalog files. Every copy matches its source. Full GET of the raw archive, export and external manifest matches, and internal/external manifests are byte equal. Raw files were ignored only after this closure. Receipts, unified export, run reports and the reproduction guide retain the commands, limits and failed attempts. The export uses one primary analysis invocation with 21 reused packets; the verifier/controls are not new historical samples. Export Schema/file admission ran; the custom plan is not canonical experiment-protocol-checker certification.

Checked the canonical IC scoreboard, progress timeline, relevant curve diagrams and performance-gain pages. This count-based exclusion supplies no attack, isogeny, speed or crossover value for those graphs. Study figures and a neutral native count-cell coverage metric are updated. Hosted merge checks and review remain separate from the saved-data and proof checks; no hosted pass is claimed.